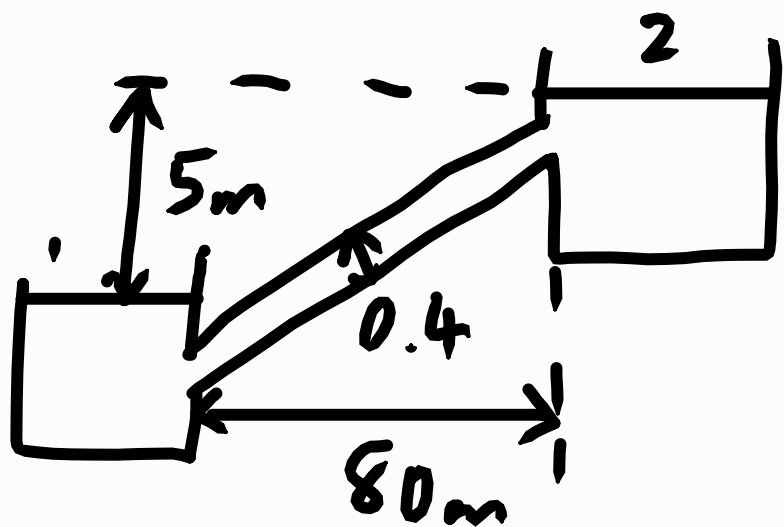


1)



Energy equation from 1 to 2:

$$h_s + \frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 - h_L = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

$$h_s = z_2 - z_1 + h_L$$

$$= 5 + h_L$$

$$\frac{V^2}{2g} = \frac{Q^2}{2gA^2}$$

$$= \frac{Q^2}{2g\left(\frac{\pi D^2}{4}\right)^2}$$

$$= \frac{8Q^2}{g\pi^2 D^4}$$

$$h_L = f \frac{L}{D} \frac{V^2}{2g} + \sum K_L \frac{V^2}{2g}$$

$$= \frac{V^2}{2g} \left(f \frac{L}{D} + \sum K_L \right)$$

$$= \frac{8Q^2}{g\pi^2 D^4} \left(f \frac{L}{D} + \sum K_L \right)$$

$$1) h_L = \frac{8Q^2}{g\pi^2 D^5} \left(f \frac{L}{D} + \sum K_L \right)$$

$$= \frac{8Q^2}{9.81\pi^2 (0.4)^5} \left(0.032 \left(\frac{80}{0.4} \right) + 4 \right)$$

$$= 33.56716074 Q^2$$

$$\approx 33.57 Q^2$$

$$h_s = 5 + 33.57 Q^2$$

Operating pump head:

$$h_s = h_p$$

$$5 + 33.57 Q^2 = 15 - 100 Q^2$$

$$Q = 0.2736214581$$

$$\approx 0.2736 \text{ m}^3 \text{ s}^{-1}$$

$$h_p = 15 - 100(0.2736)^2$$

$$= 7.513529766$$

$$\approx 7.513 \text{ m}$$

$$P = \frac{\rho g h_L Q}{\eta}$$

$$= \frac{1.08(1000)(9.81)(7.513)(0.2736)}{80\%}$$

$$= 27225.37176 \text{ W}$$

$$\approx 27.2 \text{ kW}$$

1) When $z_2 = 20$,

$$h_s = 20 + 33.57Q^2$$

$$h_s = h_p$$

$$20 + 33.57Q^2 = 15 - 100Q^2$$

$$Q^2 = -0.03743435117 < 0$$

$\therefore$ The same pump cannot work as it does not provide sufficient head.

Using 2 pumps in series:

$$20 + 33.57Q^2 = 2(15 - 100Q^2)$$

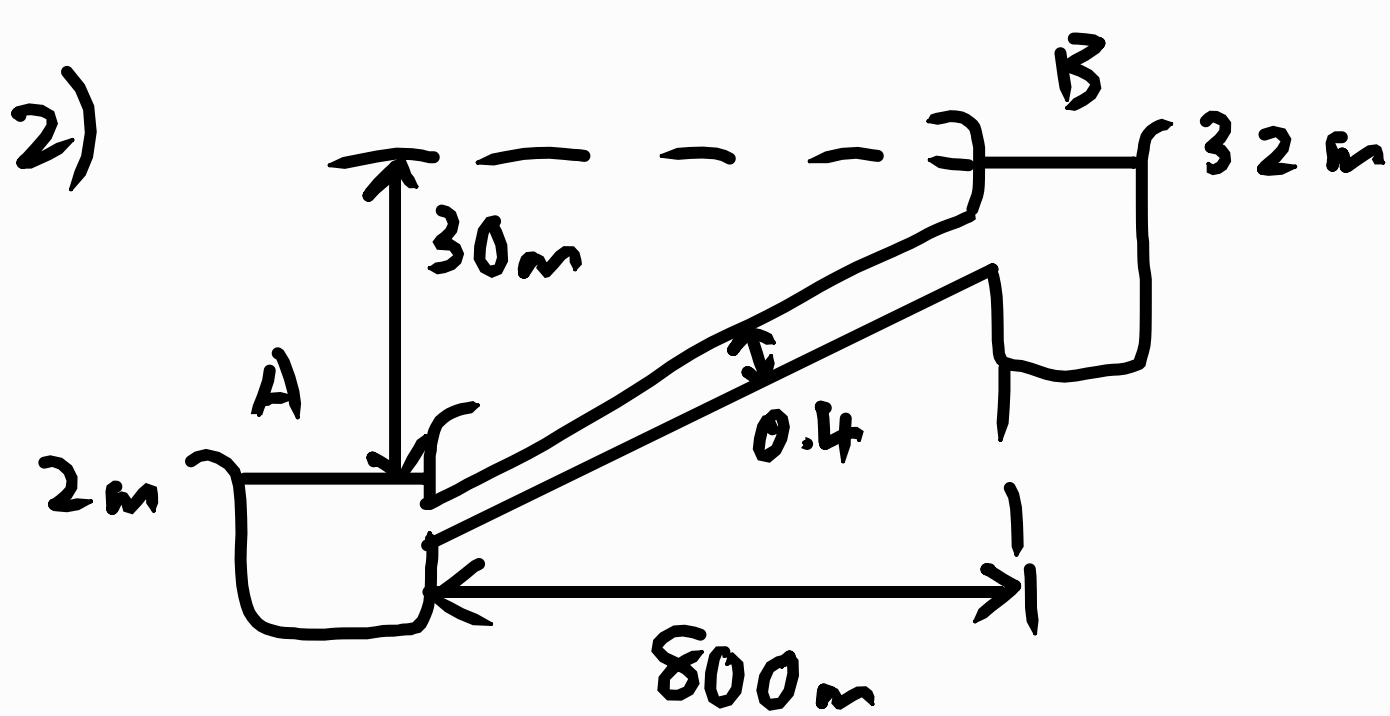
$$Q = 0.2069160167$$

$$\approx 0.207 \text{ m}^3 \text{ s}^{-1}$$

$$h_p = 2(15 - 100(0.207)^2)$$

$$= 21.43715241$$

$$\approx 21.44 \text{ m}$$

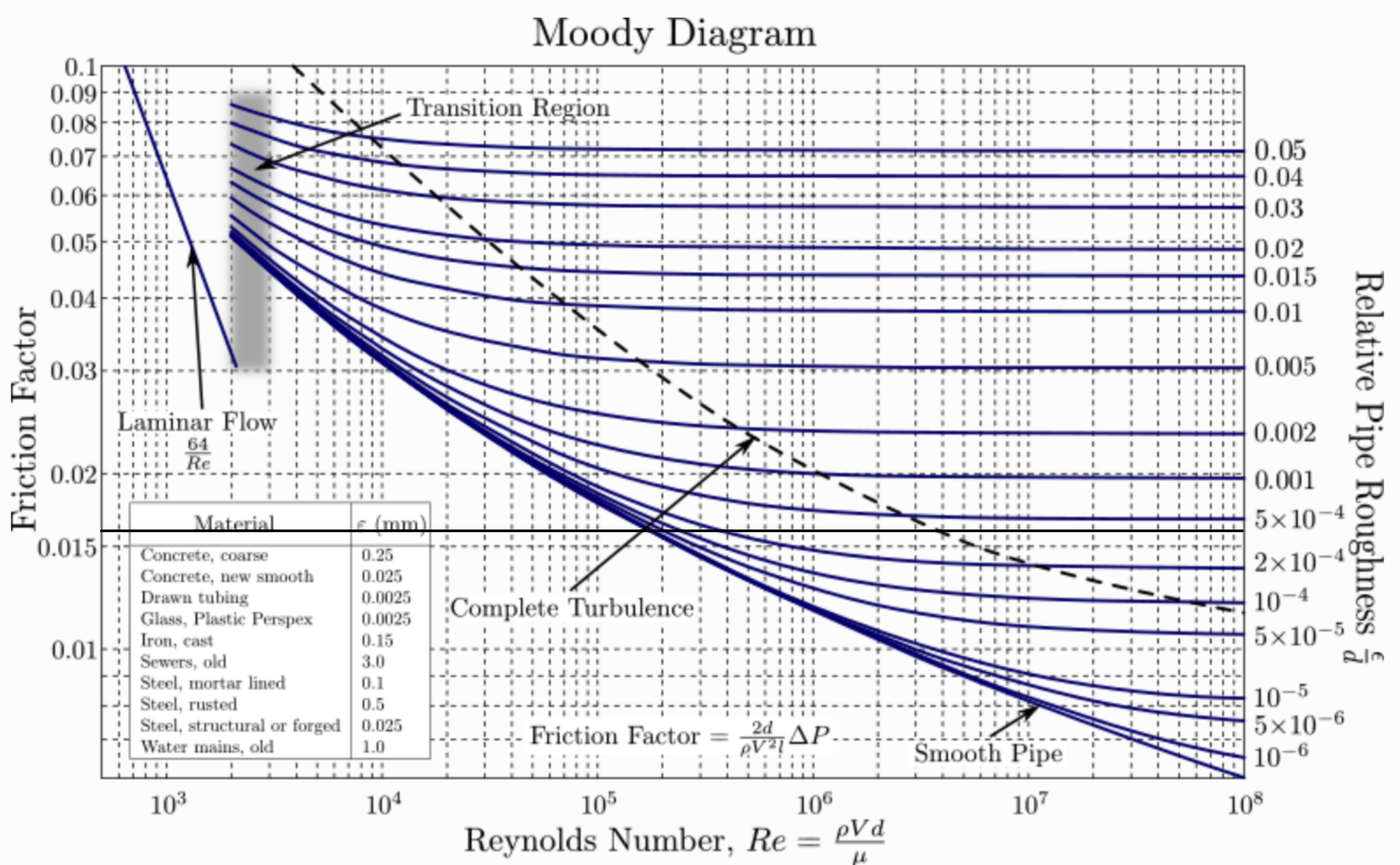


Energy equation from A to B:

$$h_p + \frac{P_A}{\rho g} + \frac{V_A^2}{2g} + z_A - h_L = \frac{P_B}{\rho g} + \frac{V_B^2}{2g} + z_B$$

$$\begin{aligned} h_p &= z_B - z_A + h_L \\ &= 32 - 2 + h_L \\ &= 30 + h_L \end{aligned}$$

$$\frac{\epsilon}{D} = \frac{0.16 \times 10^{-3}}{0.4} = 4 \times 10^{-4}$$



From the Moody chart, $f = 0.016$.

$$2) \frac{V^2}{2g} = \frac{Q^2}{2gA^2} = \frac{Q^2}{2g(\pi \frac{D^2}{4})^2} = \frac{8Q^2}{g\pi^2 D^4}$$

$$h_L = f \frac{L}{D} \frac{V^2}{2g}$$

$$= f \frac{L}{D} \frac{8Q^2}{g\pi^2 D^4}$$

$$= 0.016 \left(\frac{800}{0.4} \right) \left(\frac{8Q^2}{9.81\pi^2 (0.4)^4} \right)$$

$$= 103.2835715Q^2$$

$$\approx 103.28Q^2$$

$$h_s = 30 + 103.28Q^2$$

When $h_s = 0$, and the flow is from z_B to z_A :

$$0 = -30 + 103.28Q^2$$

$$Q = 0.5389456907$$

$$\approx 0.539 \text{ m}^3 \text{ s}^{-1}$$

$$h_p = h_s$$

$$40 - 24Q^2 = 30 + 103.28Q^2$$

$$Q = 0.2802940144$$

$$\approx 0.28 \text{ m}^3 \text{ s}^{-1}$$

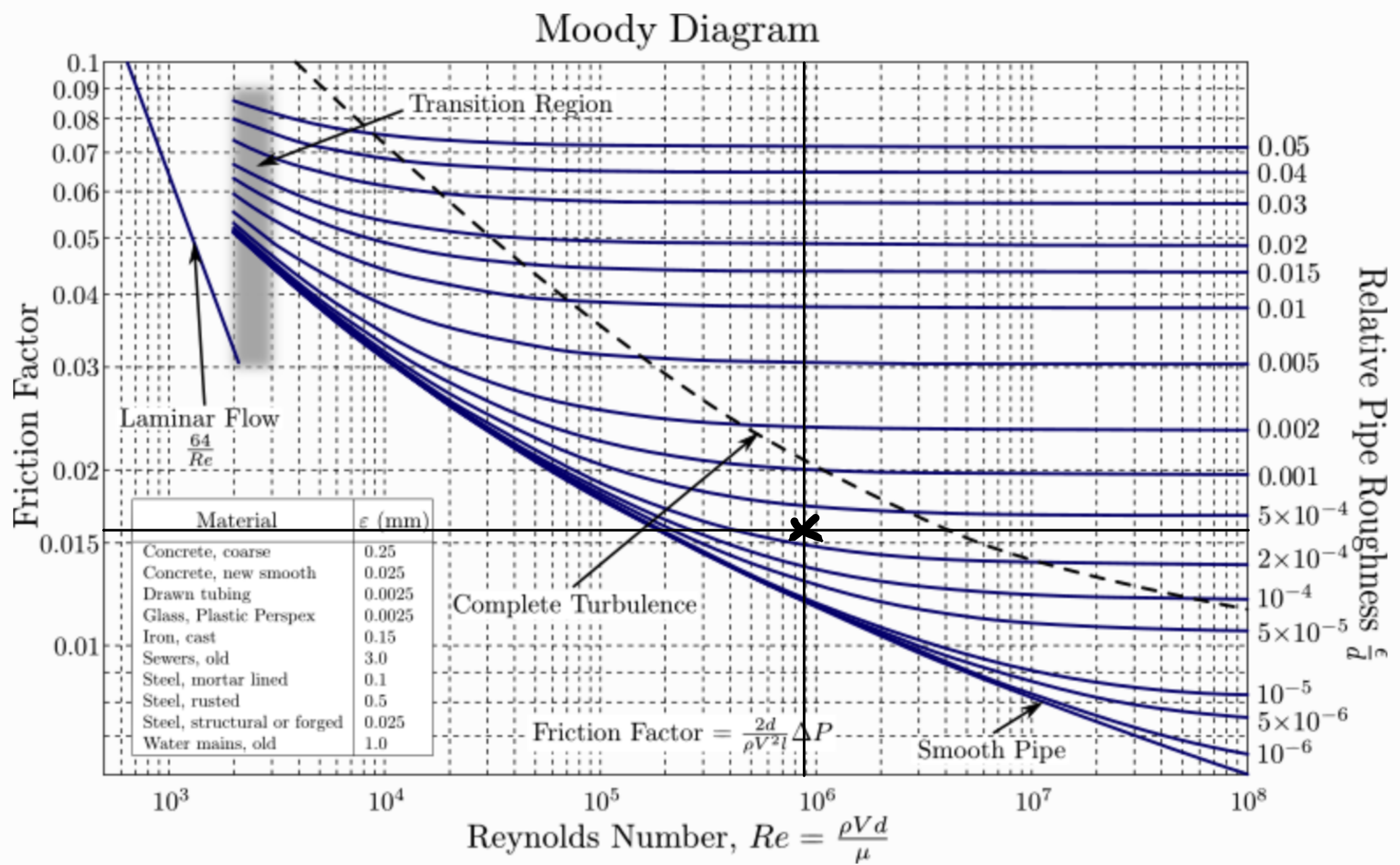
$$\begin{aligned}
 2) \quad h_p &= 40 - 24(0.28)^2 \\
 &= 38.11444637 \\
 &\approx 38.11 \text{ m}
 \end{aligned}$$

$$\begin{aligned}
 P &= \frac{\rho g h_p Q}{\eta} \\
 &= \frac{1000(9.81)(38.11)(0.28)}{0.75}
 \end{aligned}$$

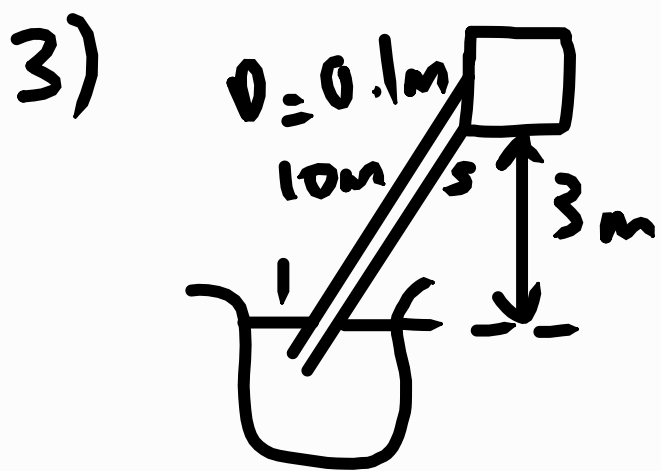
$$\begin{aligned}
 &= 139736.9254 \text{ W} \\
 &\approx 139.7 \text{ kW}
 \end{aligned}$$

$$\begin{aligned}
 Re &= \frac{\rho V D}{\mu} = \frac{\rho Q D}{\mu A} = \frac{\rho Q D}{\mu \frac{\pi D^2}{4}} \\
 &= \frac{4 \rho Q}{\pi \mu D} \\
 &= \frac{4(1000)(0.28)}{\pi(1 \times 10^{-3})(0.4)} \\
 &= 892203.5581 \\
 &\approx 8.9 \times 10^5
 \end{aligned}$$

2)



From the Moody chart, the flow is close to fully turbulent.



$$NPSH = \frac{P_s - P_v}{\rho g} + \frac{V_s^2}{2g}$$

Energy equation from 1 to s:

$$\frac{P_1}{\rho g} + \cancel{\frac{V_1^2}{2g}} + z_1 - h_L = \frac{P_s}{\rho g} + \frac{V_s^2}{2g} + z_s$$

$$\frac{P_1 - P_v}{\rho g} + z_1 - h_L - z_s = \frac{P_s - P_v}{\rho g} + \frac{V_s^2}{2g}$$

$$\begin{aligned} NPSH &= \frac{P_{atm} - P_v}{\rho g} + z_1 - z_s - h_L \\ &= \frac{101 \times 10^3 - 2340}{1000 \times 9.81} + 0 - 3 - h_L \\ &= \frac{6923}{981} - h_L \end{aligned}$$

$$\begin{aligned} h_L &= f \frac{L}{D} \frac{V^2}{2g} + K \frac{V^2}{2g} \\ &= \frac{V^2}{2g} \left(f \frac{L}{D} + K \right) \\ &= \frac{4^2}{2 \times 9.81} \left(0.015 \left(\frac{10}{0.1} \right) + 2.5 \right) \\ &= 3.261977574 \\ &\approx 3.26 \end{aligned}$$

$$\begin{aligned}
 3) \text{ NPSH} &= \frac{6923}{981} - 3.26 \\
 &= \frac{1241}{327} \\
 &\approx 3.795 \text{ m}
 \end{aligned}$$

Head coefficient:

$$C_H = \frac{gH}{\omega^2 D^2}$$

$$H = \frac{C_H \omega^2 D^2}{g}$$

$$\frac{\text{NPSH}_2}{\text{NPSH}_1} = \frac{\frac{\cancel{C_H} \omega^2 D_2^2}{g}}{\frac{\cancel{C_H} \omega^2 D_1^2}{g}} = \left(\frac{D_2}{D_1} \right)^2$$

Hence, a larger pump gives rise to a larger NPSH.

$$4) P_{atm} = 101 \times 10^3 \text{ Pa}$$

$$P_v = 1700 \text{ Pa}$$

Energy equation:

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 - h_L = \frac{P_s}{\rho g} + \frac{V_s^2}{2g} + z_s$$

$$\frac{P_s - P_v}{\rho g} + \frac{V_s^2}{2g} = \frac{P_{atm} - P_v}{\rho g} - z - h_L$$

$$NPSH = \frac{P_{atm} - P_v}{\rho g} - z - h_L$$

$$8 = \frac{101 \times 10^3 - 1700}{1000 \times 9.81} - z - 1$$

$$8 = \frac{2983}{327} - z$$

$$z = \frac{2983}{327} - 8$$

$$= 1.122324159$$

$$\approx 1.122 \text{ m}$$

If the pump is located at 2 m above the suction level, cavitation occurs.